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A Multi-Resolution Deep Learning Framework for Seizure Detection in Multi-Channel Electroencephalography

Master's Thesis

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Author's Declaration of Originality

I hereby certify that I am the sole author of this thesis. All the used materials, references to the literature and the work of others have been referred to. This thesis has not been presented for examination anywhere else.

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15.03.2026

Use of AI in Thesis

I have utilised AI tools in the preparation of this thesis.

Use of AI tools: ☐ No ☒ Yes

AI Tools Utilized

The following AI applications were employed during various stages of thesis development:

- **Claude Haiku (version 3.5)** by Anthropic
- **Gemini (version 3)** by Google

Purpose and Application of AI Tools

AI tools were utilized strategically throughout the thesis development process to enhance both technical and written aspects of the work:

Writing Quality Improvement

Gemini was employed to improve the overall quality of the thesis text by rephrasing sentences for improved clarity and academic tone, enhancing paragraph cohesion and coherence, correcting grammatical errors, and inconsistencies. Lastly, refining the overall readability of technical sections.

Technical and Programming Assistance

Claude Haiku provided technical support during implementation by debugging Python code during model development. Assisting with LaTeX document structure and formatting, providing guidance on software architecture and implementation approaches, and finally, helping with documentation and code comments. I acknowledge the full responsibility for the entire content of this thesis, including all sections where AI tools were used. I confirm that the use of AI tools has been disclosed transparently and that all AI-assisted content represents a legitimate tool for improving communication and technical quality, not a substitute for original scholarly work.

Abstract

Title: A Multi-Resolution Deep Learning Framework for Seizure Detection in Multi-Channel Electroencephalography

Epilepsy is a chronic neurological disorder characterized by a lasting predisposition to generate seizures, affecting approximately 50 million people worldwide across all socio-demographic boundaries. Despite significant progress in neurophysiological monitoring, the reliable and timely detection of ictal events remains a clinical challenge, underscoring the critical need for rapid, accurate, and noninvasive automated computational methods.

This thesis investigates whether the integration of multiple temporal resolutions through a decision-fusion architecture enhances the performance of automatic seizure detection. The proposed framework independently analyzes electroencephalography (EEG) segments of varying epoch lengths and subsequently fuses their predictive outputs to formulate a final diagnostic decision. To evaluate this multi-resolution strategy, the study utilizes the CHB-MIT Scalp EEG database. The raw signals undergo a rigorous preprocessing pipeline, including a 4th-order Butterworth band-pass filter (0.5–60 Hz), IIR notch filtering at 60 Hz to suppress power-line interference, and channel-wise Z-score normalization. To address the inherent class imbalance of the dataset, a two-stage undersampling strategy is implemented, resulting in a balanced distribution of ictal and interictal samples.

The effectiveness of the multi-resolution approach is systematically tested against fixed-epoch-length deep learning models and traditional feature-engineering pipelines. Performance is validated using a Leave-One-Subject-Out (LOSO) cross-validation scheme to ensure the clinical generalizability of the results across heterogeneous patient profiles. EEG-based decisions are directly fused to obtain a final decision. The decision fusion system adopts an override option that allows for the EEG-based decision to override the fusion-based decision in the event that the detector observes a string of EEG-based seizure decisions. The proposed detectors exhibit an improved performance, with respect to sensitivity and detection latency, compared with the state-of-the-art detectors. Experimental results demonstrate that the weighted soft voting with a 1s or 2s window remains the "Gold Standard" configuration. It provides the highest sensitivity and F1-score for the majority of patients by allowing the most reliable models to guide the final probability.

Keywords: Epilepsy, Electroencephalography, Deep Learning, Seizure Event Detection, Cross-Subject Analysis

List of Abbreviations and Terms

AI	Artificial Intelligence
ANN	Artificial Neural Networks
BiLSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
DL	Deep Learning
EEG	Electroencephalography
EMG	Electromyography
GTC	Generalized Tonic-Clonic
LSTM	Long Short-Term Memory
LOSO	Leave-One-Subject-Out
ML	Machine Learning
RNN	Recurrent Neural Network

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1 Introduction

The brain is a complex organ composed of various interconnected regions that control specific functions. It is divided into several core structures, each with distinct roles in cognition, emotion, and survival. The cerebrum is responsible for higher cognitive functions such as reasoning, problem-solving, and creativity. It governs voluntary movements and processes sensory information to help us navigate the world. The cerebrum is divided into two hemispheres that are each responsible for different cognitive abilities: The left hemisphere is often associated with logic, analytical thinking, and language processing, while the right hemisphere is linked to creativity, intuition, and spatial awareness. The cerebral cortex (the outer layer of the cerebrum) is responsible for complex thought and is related to decision-making.

The limbic system, emotions, and memory: The limbic system, including the amygdala and hippocampus, plays a critical role in emotional processing and memory formation. The amygdala helps regulate fear and pleasure responses, shaping how we react to stressful or rewarding situations. The hippocampus is crucial for forming new memories and connecting emotions to experiences. Damage to the hippocampus can result in memory impairment and difficulty learning new information. The limbic system's interactions with other brain regions determine how we process emotions and recall past experiences.

The brainstem vital functions and survival: The brainstem controls essential functions such as breathing, heart rate, and sleep. It serves as the communication highway between the brain and the rest of the body, ensuring that vital signals are transmitted efficiently. The brainstem consists of the midbrain, pons, and medulla oblongata, each with its own role in regulating involuntary processes like reflexes, digestion, and cardiovascular function. Because of its fundamental role in survival, damage to the brainstem can have life-threatening consequences (Kandel et al., 2013).

1.1 Background and Motivation

In particular, the human brain has a structure that is more difficult to decipher than the brains of other living things. To date, many studies have been conducted with various methods to understand the dynamic structure of the human brain. One of these methods is the electroencephalography (EEG) method, which investigates the electrical discharges recorded from the scalp and helps to understand brain dynamics with a high temporal resolution. EEG can be used both in neuroscience research and in neurology and psychiatric clinics to aid diagnosis and obtain information about the course of the disease (Yener & Öz, 2022).

EEG oscillations are caused by the superposition of delta (0.5-3.5 Hz), theta (4-7 Hz), alpha (8-13 Hz), beta (15-30 Hz), and gamma (28-100 Hz) waves, which have different frequency

ranges. It is formed (by overlapping waves). Brain cells communicate with each other by producing tiny electrical signals, called impulses. An EEG measures this activity, and it can be used to monitor or diagnose epilepsy and seizures.

The electroencephalographic (EEG) signal represents bioelectric potentials generated by brain activity, recorded from the scalp using electrodes and specialized equipment. EEG activity reflects the temporal summation of synchronous firing from millions of spatially aligned cortical neurons. The measurement system captures weak electrical signals from the scalp, amplifies them, processes them, including digitization, and records the resulting data. Background oscillations are often steady, while transients, sudden spikes, or shifts, can signal critical events like seizures. Yet, in hospitals around the world, the vast volume of EEG data often overtakes the speed of human analysis, making it challenging for patients and clinicians (Cascella & Bandyopadhyay, 2023).

The main characteristics of an adult EEG (Abhang et al., 2016):

Table 1.1: Main Characteristics of an Adult EEG

Frequency Band	Frequency & Brain States	Reactivity
Alpha Rhythm	Steady, smooth waves (8–13 Hz), strongest at the posterior part of the scalp when relaxed and awake.	The “idle hum” of a healthy, resting mind.
Mu Rhythm	Alpha-like rhythm (8–10 Hz) over the central brain; quiets when moving.	Linked to motor control; disappears during movement.
Beta Activity	Fast, low-voltage waves (13–40 Hz) during focus, alertness, or anxiety.	Excessive activity can signal stress, medication effects, or other influences.
Theta Waves	Slow waves (5–7 Hz), often found in the frontal region of the scalp.	Normal in children; in adults, it may indicate drowsiness or underlying issues.

Figure 1. Example of EEG waveforms showing common rhythms in a healthy adult brain. From the slow, smooth alpha waves to the rapid beta bursts. Each rhythm reflects a different state of brain activity, helping clinicians spot whether there are any abnormal patterns in the signal.

Analyzing and interpreting EEG tracings involves a combination of art and science. Normal EEGs exhibit high diversity and a broad range of physiological variability. Advanced processing techniques, such as Fourier analysis, joint time-frequency analysis, and specialized algorithms, expand the applications of EEG to general anesthesia and research areas, including neuroscience and cognitive psychology. A systematic approach to interpreting EEG waveforms is essential. Potential confounding factors, including the patient’s age, state of consciousness, physical and mental activity, and the influence of biological, environmental, and pharmacological factors on the waveforms, must be considered before analysis begins.

Before any meaningful interpretation can happen, clinicians must separate genuine brain signals from artifacts, electrical patterns generated by muscle activity or eye movement rather than the brain itself. Common Physiological Artifacts, for example: Muscle activity from the scalp can create low-amplitude, high-frequency noise, often strongest at the front of the head.

Eye movements appear as rapid deflections in the frontal regions. Blinks, in particular, produce large diphasic waves because the eye acts as an electrical dipole; when the cornea tilts upward during a blink, it creates a distinct positive–negative shift.

Swallowing or tongue movement can also generate slow waves mixed with muscle artifacts. Artifacts are common during the wakeful EEG, and one of the challenges of EEG interpretation is distinguishing these from the cerebral signal. When neurologists interpret an EEG, one of the first objectives is to assess the background activity. To examine it, clinicians view the EEG through different montages that highlight how brain signals flow across areas. This enables them to measure dominant frequencies, amplitude, and synchrony.

Abnormal events are patterns in the EEG that deviate from this baseline rhythm and suggest an abnormality. Examples include:

Table 1.2: Abnormal EEG Event Patterns (Tatum, 2016)

Event Type	Description
Spikes & Sharp Waves	Sudden, brief bursts of high-voltage activity, often linked to epilepsy.
Seizure Activity	Rapid, repetitive waveforms that spread across brain regions, seen during an actual seizure (ictal phase).
Pre-ictal Changes	Subtle shifts in rhythm or bursts of activity that happen just before a seizure begins.
Focal Slowing	Slower waves in one brain region, which may point to injury, tumor, or stroke.
Generalized Slowing	A slowdown across the whole brain, often seen in encephalopathy or coma.

These events differ from the “normal” background and can last just seconds. Without careful review or automated assistance, an EEG abnormality can be easily missed. Brain activity in individuals with epilepsy can be separated into four states: regular brain activity (interictal), brain activity before the seizure (preictal), brain activity during the seizure (ictal), and brain activity immediately after a seizure (postictal).

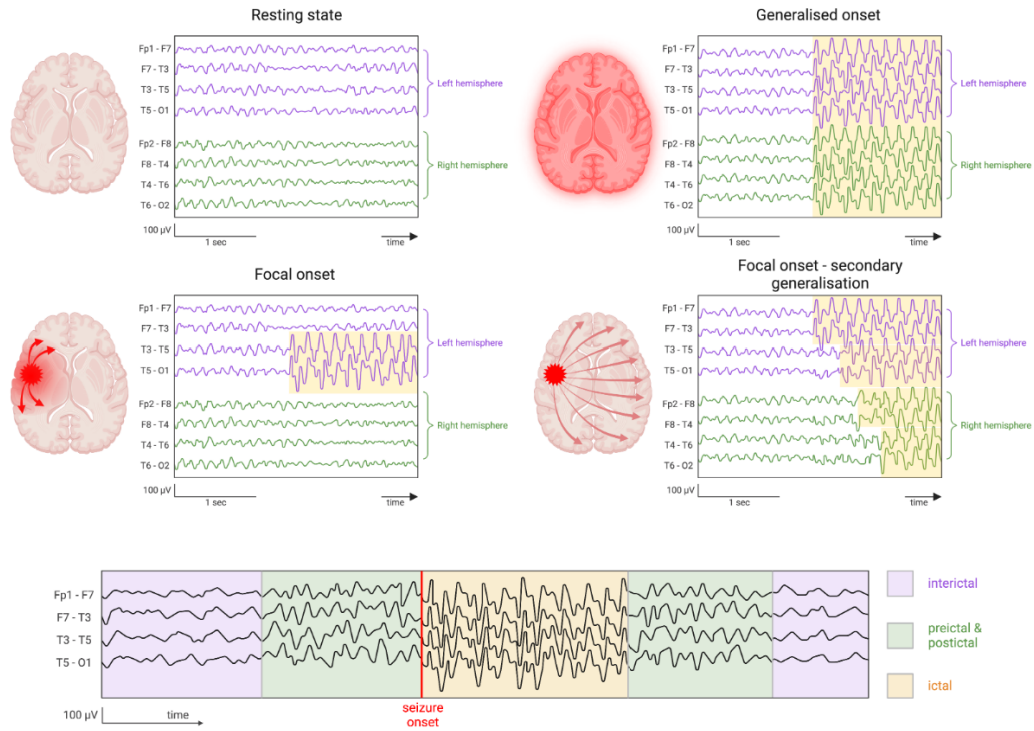


Figure 1.1: Main seizure types and some EEG characteristics

Differentiating normal waveforms from abnormal patterns is essential for EEG to achieve high specificity and diagnostic value. EEG waveforms can be characterized by location, amplitude, frequency, morphology, continuity (rhythmic, intermittent, or continuous), synchrony, symmetry, and reactivity. Frequency remains the most commonly used method for classifying EEG waveforms, with waves named according to their frequency ranges using Greek letters. Key waveforms include δ (0.5-4 Hz), θ (4-7 Hz), α (8-12 Hz), σ (12-16 Hz), and β (13-30 Hz) (Abhang et al., 2016).

Electroencephalogram (EEG)

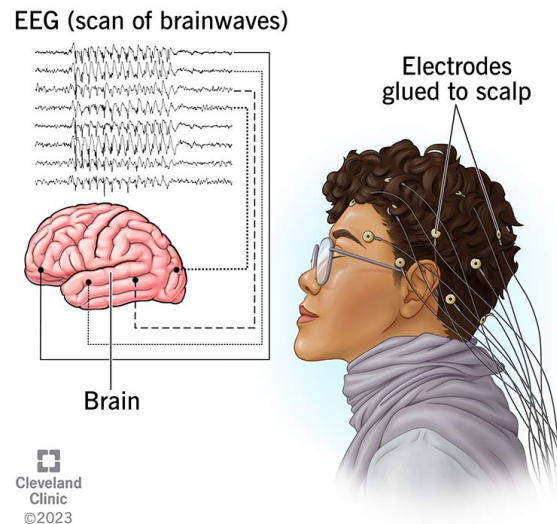


Figure 1.2: EEG visually represented (my.clevelandclinic.org).

Normal rhythms are categorized into awake and sleep patterns. Awake patterns include the rhythm and its variants, rhythm, waves, posterior slow waves of youth, hyperventilation-induced slowing, photic driving, and photomyogenic response. Sleep patterns include positive occipital sharp transients of sleep, vertex waves, spindles, K complexes, sleep-related hypersynchrony, and frontal arousal rhythm. Breach rhythms can affect both awake and sleep patterns. Additional waveforms exist outside the conventional clinical EEG bandwidth, but they have gained importance with the advent of digital signal processing. These patterns include infraslow oscillations (<0.5 Hz), waves (30-80 Hz), and high-frequency oscillations (>80 Hz). Several patterns or variants of uncertain clinical significance previously considered abnormal in early EEG history are now classified as normal.

These patterns include:

- Wicket spikes and wicket rhythms: The most commonly overread normal pattern for epileptiform activity
- Small sharp spikes: Also known as benign epileptiform transients of sleep
- Rhythmic midtemporal of drowsiness: Also called psychomotor variant
- Cigánek rhythm (midline)
- 6-Hz phantom spike-wave
- 14- and 6-Hz positive spikes
- Subclinical rhythmic epileptiform discharges of adults

- Slow-fused transients
- Occipital spikes associated with blindness
- Temporal slowing in older adults (Amin et al., 2023)

1.2 Problem Statement

EEG tracings are essential for identifying and characterizing epileptiform and other pathological patterns associated with various neurological conditions. Clinicians develop proficiency in recognizing normal EEG variations before interpreting abnormal findings. However, a single normal EEG does not rule out underlying pathology, as electroencephalographic abnormalities may be transient. Over-interpreting benign variants of uncertain significance can lead to unnecessary concern, further testing, and possible misdiagnoses. Optimal patient outcomes depend on a collaborative approach involving clinicians, nurses, and advanced practice providers who are properly trained in EEG interpretation and subsequent management.

EEG interpretation must be equally skilled in identifying benign EEG variants and in detecting pathological patterns. Neurologists, in particular, should refine their ability to distinguish between a normal variant and true epileptiform activity to ensure diagnostic accuracy and guide appropriate treatment (Sheng et al., 2024).

However, utilizing deep learning-based models in the medical domain, especially when detecting seizures has some complications. As a result of the subject-dependent and complex nature of seizures, an extreme data imbalance, complex interpretation, and the need for large amounts of training data to fine-tune hyperparameters are observed.

1.3 Research Objectives and Questions

The purpose of this paper is to argue the efficacy of deep learning models in improving the accuracy and reducing the latency of epileptic seizure detection, using the CHB-MIT Scalp EEG Dataset as a benchmark for comparison with traditional feature-engineering-based approaches. Whether by developing and evaluating lightweight or optimized deep learning architectures that maintain state-of-the-art performance while minimizing computational overhead, enabling real-time or edge deployment in clinical environments. Lastly, to evaluate the performance of a multi-resolution deep learning strategy against fixed-epoch-length models and traditional feature-engineered pipelines, focusing on metrics such as accuracy, sensitivity, and specificity for EEG-based seizure detection. The following research questions guide this investigation:

- **RQ1:** How can a deep learning model improve the accuracy and latency of epileptic seizure detection compared to traditional feature-engineering-based approaches using the CHB-MIT Scalp EEG Dataset?

- **RQ2:** Can a lightweight or optimized deep learning architecture achieve comparable performance to state-of-the-art models while reducing computational cost, making it feasible for real-time or edge deployment in clinical settings?
- **RQ3:** How does a multi-resolution deep learning strategy compare to fixed-epoch-length models and traditional feature-engineered pipelines in terms of accuracy, sensitivity, and specificity for EEG-based seizure detection?

1.4 Thesis Structure

The remainder of this thesis is organized as follows:

Chapter 2: Reviews existing literature and theoretical background.

Chapter 3: Describes the research methodology and approach.

Chapter 4: Presents and analyzes the results.

Chapter 5: Concludes the thesis and suggests future work.

The following tools and libraries were utilized:

Table 1.3: Tools and Libraries Used

Tool/Libraries	Version	Purpose
Python	3.9+	Implementation and scripting
TensorFlow/Keras	2.17+	Deep learning framework
NumPy	1.20+	Numerical computing
SciPy	1.7+	Signal processing
MNE-Python	1.0+	EEG-specific processing

2 Literature Review and Theoretical Background

Scalp EEG typically uses about 20 electrodes affixed across the skin in standardised locations, typically using a cap equipped with Ag-AgCl electrodes. A widely used international system of electrode placement and naming is the 10-20 system, which refers to the distances between adjacent electrodes being either 10 or 20 percent of the total distance across the skull, based on anatomical landmarks (see Figure 2.1). A deficiency of the 10-20 system is the inadequate coverage of the lower part of the brain, below the circumferential line formed by the electrodes Fp2, F8, T4, T6, O2, O1, T5, T3, F7, Fp1.

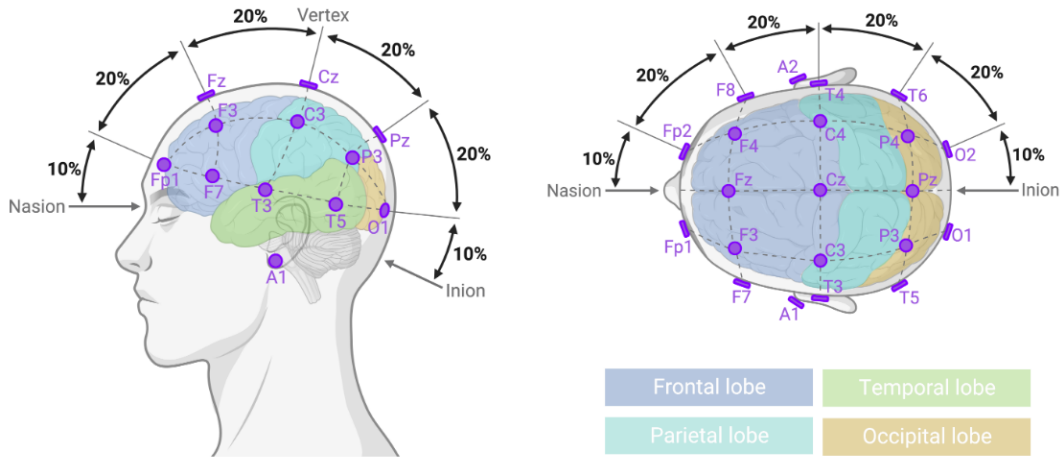


Figure 2.1: Scalp EEG

EEG signals exhibit non-linear characteristics, which pose challenges in their mathematical description, as indicated by previous studies, Pereda et al. (1998), Elger et al. (2000). Furthermore, a non-linear relationship exists between the object of interest and EEG data, which standard algorithms struggle to capture, Rakhmatulin et al. (2024). CNNs, first introduced in 1980, can learn both feature extraction and classification layers within the network. This dual learning enhances the network's generalization ability, making CNNs particularly effective for processing complex signals. Given the non-linear and intricate nature of EEG signals, CNNs are well-suited for their analysis Hassan et al. (2025).

2.1 Theoretical Background

The theoretical background of this current study is grounded in the principles of signal processing and machine learning, specifically addressing the implementation of a subject-isolated, multi-scale fusion approach to ensure both technical robustness and clinical relevance. This framework

is built upon three primary pillars: the non-stationary nature of neural dynamics, the avoidance of data leakage through segment-wise isolation, and the necessity of cross-subject validation.

2.1.1 Neural Non-Stationarity and Spatio-Temporal Dynamics

EEG signals are fundamentally characterized by high non-stationarity, meaning their statistical properties, mean, variance, and frequency spectrum change over time. Theoretically, a seizure is not a static state but a dynamic transition. Our framework utilizes a multi-scale temporal approach, employing varying window lengths ($1s$, $2s$, $4s$, $5s$, $10s$). This is based on the theory that different seizure morphologies, such as rhythmic spikes or low-voltage fast activity, manifest differently across temporal scales. By using a decision-fusion architecture, we hypothesize that the model can capture both the high-frequency "onsets" and the lower-frequency "evolutions" of a seizure event.

2.1.2 CNN

CNNs are comprised of an input layer, convolutional and pooling layers, fully connected layers, and an output layer. Similar to the convolutional layer, the pooling operation sweeps a filter across the entire input, but the difference is that this filter does not have any weights. Instead, the kernel applies an aggregation function to the values within the receptive field, populating the output array. This architecture allows CNNs to be both efficient in computation and effective in training due to the localized and globally shared connections between neurons in the convolutional layers, (X. Zhang et al., 2024).

Figure 3.3 illustrates a schematic diagram of a basic CNN architecture, which is particularly well-suited for processing structured grid-like data such as images or time-frequency representations (e.g., spectrograms). The architecture is composed of several key layers that sequentially transform the input into a set of increasingly abstract and informative features for classification. The process begins with the input layer, which typically receives a 2D input such as an image or spectrogram, represented as a matrix of pixel or intensity values. This input is passed to one or more convolutional layers, where small filters or kernels slide over the input matrix to extract local features like edges, patterns, and textures. These filters learn to activate on specific patterns during training, enabling the network to detect meaningful spatial features.

In recent years, the literature has shifted from handcrafted feature extraction toward sophisticated hybrid deep learning architectures designed to handle the high dimensionality of multichannel EEG. Craley et al. (2022) proposed SZTrack, a spatio-temporal model that utilizes a CNN encoder to process individual EEG channels and Recurrent Neural Networks (RNNs) to track the propagation of seizure activity. This approach emphasizes that seizure detection is not merely a classification task but a tracking problem across both brain space and time.

The CHB-MIT dataset is perhaps the most widely utilized benchmark in epilepsy research, there were several critical methodological flaws in its common application were identified.

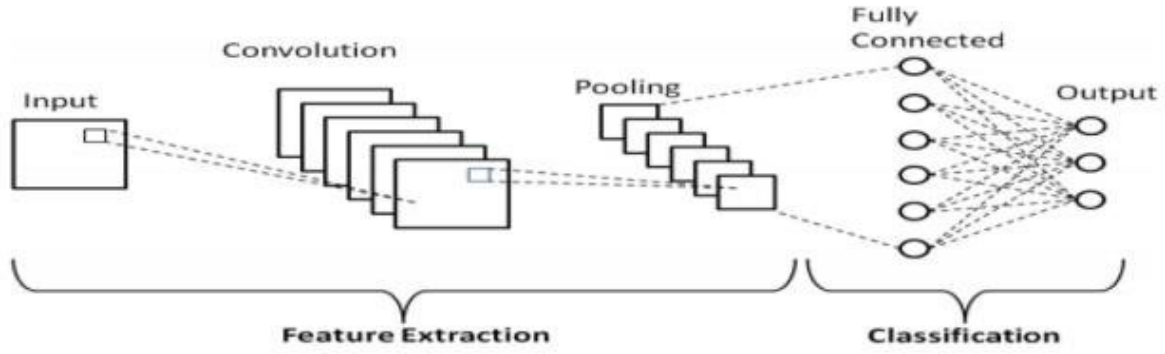


Figure 2.2: A basic diagram of a CNN architecture (Phung Rhee, 2018).

These "overlooked perspectives" suggest that many state-of-the-art results may be artificially inflated due to poor data handling practices.

A primary concern raised in the literature is the widespread use of random segment-wise splitting. In many studies, continuous EEG recordings are segmented into short epochs (e.g., 2s or 4s), which are then shuffled and randomly assigned to training or testing sets. S. Y. Shah et al. (2024) argue that this introduces severe data leakage as temporal correlation between adjacent EEG segments is exceptionally high, and the signal morphology of a single patient is highly consistent. Therefore, the model essentially "memorizes" the specific patient's EEG signature rather than learning the physiological characteristics of a seizure. When models are evaluated using this shuffled approach, they often achieve near-perfect accuracy. However, when these same models are tested on entirely new subjects (Cross-Subject or Subject-Independent settings), performance metrics often plummet. This suggests that the current body of literature may be over-reporting the effectiveness of CNNs and RNNs by ignoring the fundamental requirement for subject-level isolation during the validation phase.

Another technical gap identified in the CHB-MIT literature involves the handling of recording montages. The CHB-MIT dataset contains recordings from 22 subjects with varying channel configurations and bipolar montages. S. Y. Shah et al. (2024) highlights that many researchers overlook the necessity of channel renaming and objective bad-channel detection. Without a consistent spatial map such as the 18-common-channel structure used in this research, deep learning models receive inconsistent spatial inputs, which hampers the network's ability to learn universal spatial filters for seizure onset detection.

Historically, seizure detection has been treated as a binary classification of isolated segments. However, the clinical objective is to identify the occurrence of a seizure event from its onset to its termination.

Recent discourse has shifted toward the aggregation of segment-wise outcomes to reconstruct the seizure event. S. Y. Shah et al. (2024) emphasizes that evaluating a model on random segments does not reflect its ability to detect a seizure in real-time. If a 100-second seizure is divided into 50 segments, and the model only detects 10 of those segments correctly, but they are randomly scattered, the model has failed to provide a clinically useful alarm.

This research gap invites the exploration of multi-scale temporal windows. While shorter segments (e.g., 1s, 2s) provide high temporal resolution for pinpointing seizure onset, they are more susceptible to being triggered by technical artifacts or muscle noise. Conversely, longer windows (e.g., 10s) offer better contextual representation but may lag in detection speed. The proposal of a decision-fusion architecture, which synthesizes predictions from multiple temporal scales, addresses the non-stationarity of the EEG signal by ensuring that a detection is supported by both short-term dynamics and long-term trends.

The final "overlooked perspective" is the necessity of cross-subject generalization. The literature indicates that inter-subject variability, the difference in EEG patterns between two people, is significantly higher than intra-subject variability. Bolagh, Clifford, et al. (2017) and Nasiri and Clifford (2021) have noted that a model trained on Subject A rarely performs adequately on Subject B without significant retraining. S. Y. Shah et al. (2024) posit that for automated seizure detection to be viable in clinical practice, models must be trained and validated in a "Leave-One-Subject-Out" (LOSO) manner. This forces the neural network to extract features that are patient-agnostic, focusing on the universal spectral and spatial changes that occur during an ictogenesis, rather than the idiosyncratic background noise of a specific individual.

In summary, while the evolution from simple CNNs to complex Transformers has improved the mathematical capacity to model EEG, the literature suggests that without rigorous data isolation, proper montage standardization, and a shift toward event-based metrics, these models remain limited to theoretical scenarios. The theoretical framework of this study, therefore, directly responds to these critiques by implementing a subject-isolated, multi-scale fusion approach to ensure both technical robustness and clinical relevance.

2.2 Related Work

The emergence of hybrid deep learning models has gained popularity in the field of seizure detection. Deep Learning architectures commonly used for automated seizure detection include artificial neural networks (ANN), CNNs, and graph machine learning (GML). Different architectures classify an EEG segment based on different properties of the signal. Whilst feature-based machine learning (ML) requires expert-defined features as input, CNNs can efficiently extract local spatial features from raw EEG data without the need for hand-crafted feature extractors. This capability has made CNNs the most widely used Deep Learning technique in the field of seizure detection (Moutonnet et al., 2024).

However, despite the proliferation of high-performing models, recent critical reviews have highlighted a significant gap between reported laboratory accuracy and clinical reliability. S. Y. Shah et al. (2024) identifies several "overlooked perspectives" that suggest current benchmarks, particularly those using the CHB-MIT dataset, may be fundamentally flawed due to methodological inconsistencies. A primary concern is the widespread neglect of the non-stationary nature

of EEG signals. Most standard CNN implementations treat EEG segments as static images, failing to account for the dynamic transition from the pre-ictal to the ictal state. Furthermore, the variability in recording conditions across different subjects in the CHB-MIT dataset requires a more robust approach to data preparation than simple signal ingestion.

A critical methodological oversight identified by S. Y. Shah et al. (2024) is the issue of "data leakage" through improper validation frameworks. In many studies, researchers perform random segment-wise shuffling, where epochs from the same seizure event are split between the training and testing sets. Because temporal correlation in EEG is extremely high, the model essentially "memorizes" the specific patient's morphology rather than learning the physiological biomarkers of a seizure. This leads to what S. Y. Shah et al. (2024) describe as "near-perfect" but practically useless results, as these models fail to generalize when presented with a new, unseen patient.

Moreover, the literature often ignores the distinction between segment-level classification and event-level detection. While a CNN might correctly classify a 2-second window, clinical utility depends on the model's ability to identify a complete seizure event without excessive false alarms. This research responds to these gaps by shifting the focus toward a subject-isolated framework. By utilizing a multi-scale temporal approach incorporating windows ranging from 2 to 10 seconds, the proposed methodology aims to capture the evolving dynamics of seizure activity. This multi-resolution perspective is grounded in the theory that shorter windows provide high temporal precision for onset detection, while longer windows offer the contextual background necessary to reduce false positives caused by transient physiological artifacts.

Ultimately, the shift toward "overlooked perspectives" necessitates a move away from "black-box" accuracy and toward "Subject-Independent" (SI) models. For automated systems to be viable for real-time monitoring, they must demonstrate resilience against inter-subject variability and avoid the inflated metrics associated with random data splitting. This study integrates these critical perspectives by ensuring rigorous subject-level data persistence and a consensus-based decision-fusion architecture, providing a more transparent and clinically valid assessment of seizure detection performance.

Craley et al. (2022) proposed a neural network architecture, SZTrack, to detect and track the spatio-temporal propagation of seizure activity in multichannel EEG. SZTrack combines a convolutional neural network encoder operating on individual EEG channels with recurrent neural networks to capture the evolution of seizure activity. They evaluated SZTrack on a clinical EEG dataset of 201 seizure recordings from 34 epilepsy patients acquired at the Johns Hopkins Hospital. H. Li et al. (2024) proposed an end-to-end deep learning model by combining CNN and BiLSTM for seizure detection, which extracted features from raw EEG signals without needing time-consuming feature engineering. The model utilizes CNN to extract spatial features from multi-channel EEG signals, while BiLSTM focuses on capturing temporal dependencies of those features extracted by CNN.

ResNet, which was proposed in 2015 by researchers at Microsoft research, introduced a new architecture called Residual Network. In order to solve the problem of the vanishing/exploding

gradient, this architecture introduced the concept called Residual Blocks. In this network, we use a technique called skip connections. The skip connection connects activations of a layer to further layers by skipping some layers in between. This forms a residual block. Resnets are made by stacking these residual blocks together. In their work Zhao et al. (2024), they proposed a hybrid network model based on residual and BiLSTM (ResBiLSTM) to enhance recognition accuracy across various classification challenges. Their results reveal the superior performance of the proposed ResBiLSTM model compared to several existing hybrid methods. Although ResNet has proven powerful in many applications, one major drawback is that a deeper network usually requires weeks for training, making it practically infeasible in real-world applications. One of the most popular hybrid models is the CNN-Transformers architecture. Zhong et al. (2024) introduced a seizure detection method based on S-transform and Transformer, which used the S-transform to extract the time-frequency features of EEG signals and fed them into Transformer for feature selection and classification.

Table 2.1: Summary of reviewed studies on EEG-based epileptic seizure detection.

Study	Model / Method	Key Techniques	Dataset(s)	Performance
Fisher et al. (2014)	Clinical consensus	ILAE diagnostic criteria	N/A (clinical)	–
Tzallas et al. (2012)	Time-frequency classifier	STFT, Spectral decomposition	Bonn EEG	Accuracy: $\approx 97\%$
Shoeb et al. (2009)	SVM with multimodal fusion	EEG–ECG fusion features	CHB-MIT + ECG	Sensitivity: $\approx 96\%$
Alotaiby et al. (2014)	Channel selection survey	CSP, PCA, statistical features	Multiple	– review
Shoeb & Gutttag (2010)	Patient-specific ML	Handcrafted EEG features	CHB-MIT (introduced)	– (dataset paper)
Ali et al. (2024)	Random Forest	92 handcrafted features, 22 channels	CHB-MIT (full)	Sensitivity: 72.63% (5-fold), 75.34% (LOO)

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Study	Model / Method	Key Techniques	Dataset(s)	Performance
Roy et al. (2019)	Systematic DL review	CNN, RNN, hybrid architectures	Multiple	– (review)
Acharya et al. (2018)	13-layer CNN	Raw EEG, end-to-end learning	Bonn EEG	Accuracy: $\approx 88\%$
Raghu et al. (2025)	Systematic DL review	DL on intracranial EEG signals	Multiple (iEEG)	– review
Thodoroff et al. (2016)	CNN + LSTM	Spatiotemporal EEG features	CHB-MIT, iEEG Portal	AUC: ≈ 0.80
Daoud & Bayoumi (2019)	Conv. Autoencoder + BiLSTM	Unsupervised pre-training, sequence modelling	CHB-MIT	Accuracy: $\approx 99\%$
Ahmedt-Aristizabal et al. (2020)	Spatiotemporal CNN	Joint spectral + spatial features	CHB-MIT	Sensitivity: $\approx 97\%$
O’Shea et al. (2018)	Multi-resolution WaveNet	Dilated causal convolutions	Neonatal EEG (Helsinki)	AUC: ≈ 0.93
Usman et al. (2024)	Multi-input DL network	Parallel multi-representation inputs	CHB-MIT, Bonn	Accuracy: $\approx 99\%$
Li et al. (2024)	Hybrid DL with feature fusion	Multi-domain feature extraction	Multiple	F1: $\approx 98\%$
Zhang et al. (2025)	Multi-head attention + CNN	Spatiotemporal + attention features	CHB-MIT	Accuracy: $\approx 99\%$
Liang et al. (2025)	Multi-scale phase-aware fusion	Adaptive phase modelling across bands	CHB-MIT, Bonn	Accuracy: $> 99\%$

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Study	Model / Method	Key Techniques	Dataset(s)	Performance
Hussain et al. (2025)	Comparative DL study	CNN, attention, hybrid pipelines	Multiple	– review

The reviewed literature spans across basic clinical definitions through to state-of-the-art deep learning architectures for EEG-based seizure detection, collectively reflecting the rapid evolution of this field. (Fisher et al., 2014) established the International League Against Epilepsy (ILAE)’s practical clinical definition of epilepsy, providing the medical grounding upon which automated detection systems are designed. This consensus definition, requiring at least two unprovoked seizures or one seizure with high recurrence risk, substantiates the motivation for reliable automated systems. Traditional signal processing approaches. Early work by Tzallas et al. (Tzallas et al., 2012) demonstrated the utility of time-frequency analysis for seizure detection in EEG, achieving high classification accuracy through spectral decomposition. Shoeb and colleagues (Shoeb et al., 2009) explored multi-modal fusion of EEG and ECG signals for seizure onset detection, preluding the multi-input paradigms that dominate current literature. Alotaiby et al. (Alotaiby et al., 2014) provided a systematic overview of EEG channel selection strategies, highlighting how reducing channel dimensionality can improve classifier performance without loss of discriminative information. Deep learning reviews and CNN-based methods. Roy et al. (Y. Roy, Banville, et al., 2019) provided a systematic review of deep learning applied to EEG analysis, cataloguing architectures from CNNs to RNNs and identifying key challenges including data scarcity and inter-subject variability. Acharya et al. (Acharya et al., 2018) demonstrated that deep CNNs could automatically learn discriminative features from raw EEG for seizure detection, outperforming handcrafted feature approaches on several benchmarks. Raghu et al. (Raghu et al., 2025) extended this survey to intracranial EEG, highlighting the superior signal quality of iEEG and the architectural trends in seizure detection from implanted recordings. Recurrent and spatiotemporal models. Thodoroff et al. (Thodoroff et al., 2016) applied RNN architectures to learn spatially robust EEG representations, emphasizing the importance of temporal dynamics in seizure characterization. Daoud and Bayoumi (Daoud & Bayoumi, 2019) combined a convolutional autoencoder with a Bidirectional LSTM for seizure prediction, achieving efficient performance suitable for embedded implementations. Ahmedt-Aristizabal et al. (Ahmedt-Aristizabal & et al., 2020) proposed a deep spatiotemporal CNN that jointly captures local spectral and inter-channel spatial patterns, demonstrating improved generalization across subjects. Multi-resolution and fusion architectures, O’Shea et al. (O’Shea et al., 2018) introduced a multi-resolution WaveNet for neonatal seizure detection, showing that dilated causal convolutions can capture seizure morphology across multiple temporal scales simultaneously. Usman et al. (Usman et al., 2024) proposed a multi-input deep feature learning network that ingests multiple EEG representations in parallel, improving detection robustness.

Li et al. (X. Li et al., 2024) combined feature fusion with hybrid deep learning models for simultaneous detection and prediction, reporting strong performance on multiple public datasets. Attention and multi-scale mechanisms. Zhang et al. (Y. Zhang et al., 2025) developed a model integrating multi-head attention with spatiotemporal feature extraction, enabling the network to focus selectively on seizure-relevant EEG segments. Liang et al. (Liang et al., 2025) proposed an Adaptive Multi-Scale Phase-Aware Fusion Network that explicitly models phase relationships across frequency bands, addressing limitations of amplitude-only approaches. Hussain et al. (Hussain et al., 2025) provided a comparative study of deep learning pipelines for EEG-based seizure diagnosis, identifying attention-augmented convolutional networks as consistently top-performing. Overall observations across the reviewed studies, several recurring themes emerge: (1) deep learning consistently outperforms classical machine learning when sufficient labeled data is available; (2) cross-subject generalization remains an open challenge, with most high-accuracy results coming from subject-specific or leave-one-out designs; (3) the CHB-MIT dataset is widely used but often under-evaluated with respect to class imbalance and event-level metrics; and (4) multi-scale, attention, and fusion mechanisms represent the current frontier for both scalp and intracranial EEG seizure detection.

2.3 Research Gap

Unlike any other physiological signals, EEG signals are highly nonlinear and non-stationary (Robbins et al., 2020). The inter-subject variability in the EEG signal is much higher than the intra-subject changes (Bolagh, Clifford, et al., 2017; Nasiri & Clifford, 2021). This lack of stationarity arises from rapid transitions between neural assembly states, the presence of multiple time scales in brain activity, and varying influences from deeper brain regions.

Unlike previous studies that mostly focus on epileptic signal classification, this thesis investigates multiple temporal resolutions through a decision-fusion architecture to enhance the performance of automatic seizure detection. The proposed model analyzes EEG segments of different epoch lengths independently and fuses their predictions to form a final decision. Using the CHB-MIT dataset, the effectiveness of this multi-resolution strategy will be systematically compared with fixed-epoch-length deep learning models as well as traditional feature-engineering pipelines, enabling a thorough assessment of its relative benefits. In the current study, we specifically utilized a simple CNN model to compare the effect of different epoch lengths for epileptic seizure detection, filling a research gap in this field. The experimental results show that our model demonstrates good generalizability and reliability, making it suitable for different data environments.

Different approaches can be used to detect seizure occurrences from EEG signals. One of the most common approaches is to classify segments that contain epileptic seizures. This is achieved by dividing the continuous raw EEG signals into smaller parts, a process known as segmentation. Typically, an ML model is trained on 70–80 percent of the randomly selected

segments from the entire dataset. The remaining 20–30 percent of segments, unused during training, are used to test and evaluate the trained model. The second step is the classification of events. In this method, the event is kept at its original duration and can be isolated and detected as a whole. The full length of a seizure or non-seizure episode can be segmented out and then used in the analysis to classify it as a seizure or a non-seizure.

As an alternative to separating the entire event for detection, one can aggregate the outcomes of shorter segments. The identified segments can be reordered according to their original temporal sequence to verify whether all, or at least most, of the segments within the complete event have been correctly detected. Event detection is a more practical and realistic metric for real-world implementations, as it evaluates whether a model can identify complete seizure events rather than merely detecting arbitrary segments. However, the majority of prior work has overlooked this crucial aspect and has instead concentrated predominantly on detecting random seizure segments. Several studies have addressed event detection, encompassing both the segmentation and identification of entire events (as described earlier) and the detection of seizure onset, i.e., the initial phases of a seizure episode. Among these, only one study explicitly addressed the issues of data leakage and random segment detection by examining event detection performance under a cross-subject setting.

2.4 Summary

Most existing studies rely on segment-level classification, where continuous EEG signals are partitioned into short, isolated windows. By randomly splitting these segments into training and testing sets (e.g., a 70/80-20/30 split), models often benefit from high temporal correlation and data leakage, failing to reflect real-world clinical performance. While some studies have moved toward "Event-Based Detection," evaluating the model's ability to identify a complete seizure episode the majority of prior work continues to overlook the necessity of cross-subject validation to ensure reliability in unseen data environments.

A critical, yet under-explored, dimension in seizure detection is the impact of window size on classification performance. Most current architectures either utilize a fixed segment length (e.g., 1s or 5s) or attempt to capture multi-scale features implicitly through deep network layers. There remains a distinct lack of systematic analysis regarding the marginal gains of different epoch lengths specifically the trade-off between the rapid detection of 1-second windows versus the contextual robustness of 10-second windows.

Currently, according to our knowledge, there is no explicit decision-fusion framework applied to scalp EEG, specifically the CHB-MIT dataset, that systematically integrates multiple epoch lengths and cross-subject validation to optimize seizure detection. This study fills this gap by utilizing a streamlined CNN architecture focused on event-based detection and decision-fusion, providing a more ecologically valid metric for clinical implementation.

3 Research Methodology

The methodology employed in this study is described using the flowchart presented in Figure 3.1. The process involves several key steps, each crucial for the accurate detection of seizures using EEG data.

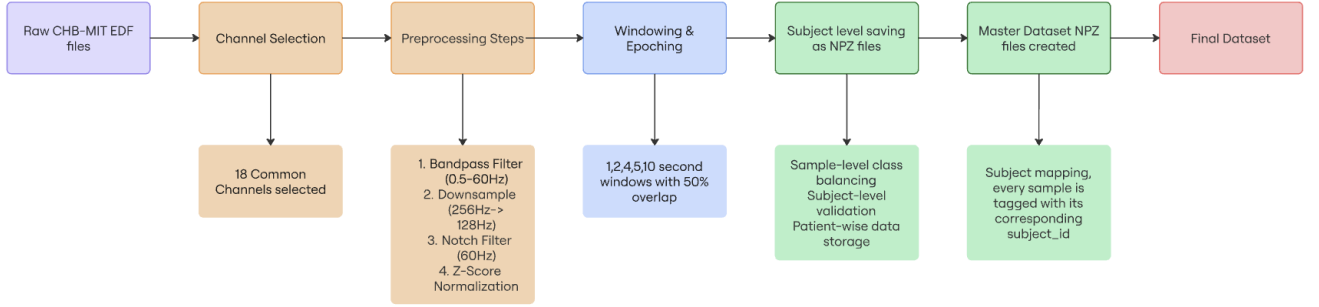


Figure 3.1: EEG Pipeline Flowchart

3.1 The CHB-MIT Scalp EEG Dataset

EEG signal abnormalities offer key insights into the brain mechanisms behind various neurological disorders. By analyzing these signals, one can gain a clear understanding of the behavioral and cognitive impacts of such conditions. The following sources describe how EEG datasets include recordings from both healthy individuals and epilepsy patients.

The study begins with the acquisition of raw EEG data in the European Data Format (EDF) from the CHB-MIT scalp EEG dataset from Physionet (<https://physionet.org/>), which was first introduced Shoeb et al. (2004). It is one of the most widely used scalp EEG datasets for seizure prediction and detection. The dataset was obtained from 24 pediatric subjects, including 23 unique patients (subject 21 was recorded 1.5 years after subject 01 from the same patient). The cohort comprised males aged 3 to 22 years and females aged 1.5 to 19 years. Each subject exhibited at least two types of seizures from three categories: partial (SP), complex partial (CP), and generalized tonic-clonic (GTC).

The seizures were located in the hippocampus (deep temporal region) in eight subjects and in the neocortex in eleven subjects. The dataset consists of 686 records, totaling over 950 hours of continuous EEG recordings. The EEG signals in this dataset were recorded using five distinct channel groups, each comprising 22 to 36 unique EEG channels. Detailed specifications of these channel configurations are provided in Table 1 of Appendix C in the electronic supplementary

material.

For this study, one channel group was selected for analysis, chosen based on its prevalence in the dataset. This group was utilized in 655 out of 686 records and includes 175 of the 198 seizure events documented. The selected group consists of 22 unique EEG channels: FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, P8-O2, FZ-CZ, CZ-PZ, P7-T7, T7-FT9, FT9-FT10, and FT10-T8. From the raw EEG data, 18 common bipolar channels were selected. These channels are chosen for their consistency across all patients, ensuring that the analysis is based on a uniform set of data points.

3.2 Data Preprocessing

Data pre-processing includes a number of possible steps (see Table 3.1). Artifacts and noise sources can disrupt the signals of the scalp EEG data in various ways. Removing or filtering out contaminated data of artifacts can therefore improve the signal-to-noise ratio and data consistency, potentially improving seizure detection performance. However, pre-processing is a two-edged sword as it contains the risk of removing informative signals like epileptiform activity. Furthermore, the run time of some pre-processing steps can reduce the potential of an algorithm to operate in real-time.

To our knowledge, a range of techniques have been applied to reduce the impact of noise in the reviewed studies; however, none of the reviewed studies benchmark and compare the effectiveness of their techniques on classification performance. Filtering techniques are applied to dilute selected frequencies in the EEG. A filter can be designed to remove frequencies above a threshold (low-pass), below a threshold (high-pass), frequencies that are not within a chosen interval (band-pass), or to remove those that are (notch). High-pass filtering is commonly used to remove the effect of the slow (< 1 Hz) nonneural baseline signal drift. A low-pass filter is often used during recording to inhibit aliasing, which occurs when the sampling frequency is lower than twice the maximum frequency in the signal. Low-pass filtering, considered to mainly contain irrelevant noise and artifacts, is also often used offline to reject frequencies above 30-45 Hz. However, this may result in the omission of informative high-frequency signals.

In practice, 0.5-40 Hz band-pass filtering is generally applied to attenuate both high-frequency and low-frequency components. In the US, 60 Hz notch filters are commonly used to remove power line interference, and 50 Hz in Europe. The preprocessing steps taken for the CHB-MIT dataset were as follows:

Step	Operation	Description
1	Downsampling	Reduced sampling rate from 256 Hz to 128 Hz.
2	Channel Renaming	18 common channel names selected; duplicate T8-P8 channels removed.
3	Band-pass Filtering	Applied a 0.5–60 Hz Butterworth filter to isolate clinical activity.
4	Epoching	Segmented into 1s, 2s, 4s, 5s, and 10s windows with 50% overlap.
5	Z-score Normalization	Standardized amplitudes to zero mean and unit variance.
6	Subject-level Data Saving	Data saved as NPZ files with class balancing and validation.
7	Master Dataset Creation	Compilation of subject files into a tagged, unified dataset.

Table 3.1: Preprocessing and Data Preparation Steps for CHB-MIT Dataset

3.2.1 Downsampling

The original CHB-MIT dataset recordings were sampled at 256 Hz. To reduce computational complexity while preserving clinically relevant information, the signals were downsampled to 128 Hz. This sampling rate is sufficient to capture the frequency content of interest for seizure detection, as most epileptiform activity occurs below 60 Hz. Downsampling was performed using an anti-aliasing low-pass filter to prevent aliasing artifacts, followed by decimation to achieve the target sampling rate.

In order to improve the quality of the EEG data, preprocessing techniques were applied (Coelli et al., 2024), which comprises preparing raw EEG signals by reducing noise, applying bandpass filtering, and epoching data to improve its disposability for classification analysis tasks. The original EEG signals exhibit high dynamics and non-stationarity, and they are often subject to interference from physiological noise, environmental noise, and 60 Hz industrial frequency signals during acquisition. In order to ensure the accuracy and reliability of subsequent EEG signal analysis, it is imperative to remove the noise in the original EEG signals and enhance the signal-to-noise ratio.

Initially, the 5th-order Butterworth band-pass filter in the frequency range of 0.5-60 Hz was used to filter the artifacts and noise from the original EEG signals. This configuration effectively removes low-frequency drift, baseline fluctuations, and certain power line interference (S. Y. Shah et al., 2024). Although CHB-MIT recordings have a common bipolar montage, channel renaming and objective bad-channel detection were applied to ensure structural consistency and robustness across all subjects and recordings.

3.2.2 Channel Renaming

Channel renaming was performed to ensure consistency across all recordings in the CHB-MIT dataset. The International 10-20 system electrode placement scheme was used as the standard nomenclature. During this process, duplicate channels were identified and removed, specifically the duplicate T8-P8 channel that appeared in some recordings. This standardization step was crucial for ensuring compatibility across different patient recordings and preventing errors during subsequent preprocessing and model training phases. Re-dimensioning the channels ensured that all EEG samples had uniform dimensions, which is a requirement for batch processing in deep learning frameworks. This step involved padding shorter recordings or truncating longer ones to achieve consistent temporal dimensions. For recordings with varying numbers of channels due to missing or excluded electrodes, zero-padding or interpolation techniques were applied to maintain a consistent channel count across all samples.

3.2.3 Band-pass Filtering

A band-pass filter with cutoff frequencies of 0.5 Hz and 60 Hz was applied to isolate the frequency range of clinical interest. The high-pass component (0.5 Hz) removed slow drifts and DC offsets, while the low-pass component (60 Hz) eliminated high-frequency noise, including powerline interference and muscle artifacts. Lastly, to attenuate unwanted AC line noise (e.g., 50 Hz in Europe, 60 Hz in North America) while preserving the rest of the EEG signal was applied. As the CHB-MIT dataset was obtained in the United States, 60 Hz power line interference from recorded brain signals was applied. This filtering step enhanced the signal-to-noise ratio and focused the analysis on frequency bands known to contain seizure-related activity.

3.2.4 Epoching

The continuous EEG recordings were segmented into non-overlapping epochs of varying durations: 1, 2, 4, 5, and 10 seconds. This multi-scale epoching approach allowed for the investigation of optimal temporal window sizes for seizure detection. Shorter epochs (1-2s) provided higher temporal resolution for detecting seizure onset, while longer epochs (5-10s) captured more contextual information about seizure evolution and background brain activity.

3.2.5 Data Normalization

Z-score normalization was applied to standardize the EEG signal amplitudes across different channels and recordings. For each channel, the mean was subtracted, and the result was divided by the standard deviation, transforming the data to have zero mean and unit variance. This normalization step was crucial for mitigating inter-subject variability in signal amplitude and ensuring that the model could generalize across different patients and recording conditions.

3.2.6 Subject-level Data Saving (NPZ)

Processed segments were saved at the subject level using the compressed numpy (NPZ) format to maintain the integrity of individual patient records. During this phase, two critical sub-steps were performed:

- **Sample-level Class Balancing:** To prevent model bias, the number of seizure and non-seizure epochs was balanced at the subject level.
- **Subject-level Validation:** Data was validated for quality and clinical relevance before final storage to ensure no technical artifacts corrupted the patient-specific files.

The individual subject-level NPZ files were combined to create a consolidated Master Dataset. Each sample in this dataset is tagged with a unique subject ID, enabling patient-wise cross-validation and easy tracking of performance metrics. The final dataset is structured to support a **decision-fusion architecture**. In this proposed framework, the CNN model analyzes segments of different epoch lengths independently, subsequently fusing their predictions to determine if multiple temporal resolutions enhance the accuracy of automatic seizure detection.

3.3 Addressing The Class Imbalance Problem

The inherent sparsity of ictal events in long-term EEG monitoring creates a severe class imbalance problem. Imbalanced data causes several significant challenges in machine learning. For instance, in an imbalanced dataset like CHB-MIT, the majority class dominates the model’s predictions, leading to high accuracy but poor performance for the minority class.

Models tend to underperform on the minority class, leading to poor recall sensitivity or precision for that class. Metrics like accuracy become misleading when working with imbalanced datasets. Precision, recall, F1-score, and confusion matrices become more relevant, as they highlight the minority class’s performance more accurately. To address this challenge, a multi-stage undersampling strategy was implemented during the preprocessing and data aggregation phases.

3.3.1 Undersampling

During the initial windowing phase, a 1-in-20 selection rule was applied to the majority class (non-seizure segments). A modulo-based filter was implemented to select interictal windows:

$$\text{selection} = \begin{cases} \text{include} & \text{if normal_counter mod } 20 = 0 \\ \text{exclude} & \text{otherwise} \end{cases}$$

This approach reduced the interictal volume by approximately 95% prior to storage, significantly lowering the memory footprint and preventing the “swamping effect” during data aggregation.

This stage operates at the file I/O level, preventing unnecessary storage of redundant non-seizure samples.

Following global aggregation of processed epochs, a final refinement of the class distribution was performed. The pipeline enforces a fixed 2:1 ratio (interictal to ictal), achieved through randomized index selection:

$$n_{\text{interictal}} = 2 \times n_{\text{ictal}} \quad (3.1)$$

A stochastic selection process using `np.random.choice` with a fixed seed extracts a representative subset from the decimated interictal pool. The resulting master dataset comprises approximately 33.3% seizure samples and 66.7% non-seizure samples. This distribution provides the model with sufficient exposure to the minority class while maintaining a realistic representation of the background EEG environment.

To ensure that the model learns generalized physiological features rather than temporal or patient-specific artifacts, the final dataset was subjected to global random permutation. This ensures that samples from the same patient or recording session are distributed stochastically across training batches, reducing the risk of temporal autocorrelation bias.

3.4 Model Training Configuration

This section details the experimental framework, optimization parameters, and validation strategies employed to develop the base CNN models across varying temporal resolutions. The models were implemented using the TensorFlow/Keras framework and optimized for execution on multi-GPU hardware. The training objective was the minimization of binary cross-entropy loss. Optimization was performed using the Adam algorithm with a fixed learning rate of 1×10^{-4} . A batch size of 128 was selected to maximize computational throughput during the training phase. While the models were trained for a maximum of 100 epochs, the final classification metrics and decision logic were evaluated at a fixed sensitivity threshold of 0.3 to account for the class imbalance inherent in seizure detection tasks.

3.4.1 Validation Strategy: Leave-One-Subject-Out (LOSO)

To ensure that the seizure detection system generalizes effectively across the biological variability present in the CHB-MIT dataset, a Leave-One-Subject-Out (LOSO) cross-validation protocol was utilized. In each fold of this 24-subject analysis, data from a single subject was entirely withheld to serve as the independent test set. Within the remaining pool of 23 subjects, a secondary subject-level split was implemented to isolate a validation set. This validation set was used exclusively for monitoring model convergence and triggering early stopping criteria, ensuring that the test set remained strictly unseen during the optimization process.

3.4.2 Training Regularization and Callbacks

To mitigate the risk of overfitting and ensure the selection of the most robust model state, several automated callbacks were integrated into the training loop:

- **Early Stopping:** A patience of 40 epochs was applied to the validation loss. Training was terminated prematurely if no improvement was observed, at which point the model weights associated with the lowest validation loss were restored.
- **Model Checkpointing:** During each epoch, the system evaluated the validation loss and automatically cached the model state (`.keras` format) only when a new global minimum was achieved.

3.5 Multi-Resolution and Decision Fusion

The proposed approach employs a decision-fusion architecture that analyzes EEG segments across multiple temporal resolutions simultaneously. Rather than committing to a fixed epoch length, the model processes epochs of 1, 2, 4, 5, and 10 seconds in parallel, with each stream extracting resolution-specific features.

Because the CHB-MIT dataset was segmented into varying epoch lengths ($L \in \{1, 2, 4, 5, 10\}$ seconds), the resulting prediction vectors were of unequal lengths. To perform decision fusion, we utilized a zero-order hold (upsampling) method to align all model outputs to the 1-second resolution baseline, ensuring that voting was performed on synchronized temporal segments.

3.5.1 Architecture Overview

The proposed system utilizes a Multiresolution Decision-Fusion (MDF) architecture designed to capture the diverse temporal characteristics of ictal (seizure) activity in EEG signals. By processing the signal across multiple temporal scales simultaneously, the model balances the need for high-resolution onset detection with the stability provided by long-term contextual windows. The high-level system design is illustrated in Figure 3.2.



Figure 3.2: Proposed Multi-Resolution Decision-Fusion Architecture. The framework processes EEG segments through five parallel streams ($1s, 2s, 4s, 5s, 10s$), followed by temporal alignment and decision fusion.

The architecture is composed of four primary functional stages:

Multi-Resolution Input Streams: The raw EEG data is segmented into five parallel streams with window lengths of $L \in \{1s, 2s, 4s, 5s, 10s\}$. This multi-scale approach ensures that both transient paroxysmal discharges (captured by shorter windows) and evolving rhythmic patterns (captured by longer windows) are represented.

Parallel Feature Extraction: Each stream feeds into an independent Convolutional Neural Network (CNN) base. While the architecture of these CNNs is identical, utilizing Global Average Pooling to handle variable input dimensions, each model is specialized to the feature density of its specific temporal resolution.

Temporal Alignment (Indexing): To facilitate decision fusion, the output probabilities from varying resolutions must be synchronized. Since a 10s window corresponds to ten 1s intervals, we apply a zero-order hold (upsampling) method to align all prediction vectors to the 1s base resolution. This ensures that the fusion mechanism operates on temporally consistent segments.

Decision Fusion Layer: The synchronized predictions are combined using an ensemble learning strategy. The system supports both *Hard Voting* (majority consensus) and *Soft Voting* (averaging of class probabilities). This stage acts as a temporal filter, effectively reducing false positives by requiring cross-resolution agreement before a seizure alarm is triggered.

By integrating these components, the architecture leverages the precision of micro-scale temporal data and the robustness of macro-scale observations, providing a comprehensive analysis of the CHB-MIT dataset.

3.5.2 Ensemble Learning

The final stage of the proposed architecture is governed by Late Fusion Theory, where decision-level information from multiple independent classifiers is aggregated to produce a singular, robust output. Rather than relying on a fixed temporal resolution, this research implements an ensemble learning framework that integrates findings from five distinct epoch lengths: 1s, 2s, 4s, 5s, and 10s.

A significant challenge in multiresolution ensemble learning is the inherent temporal misalignment of prediction vectors; for example, a 10s model produces an order of magnitude fewer predictions than a 1s model for the same duration of EEG data. To achieve a synchronized consensus, we implement a temporal alignment procedure using linear interpolation. This process upsamples the lower-resolution prediction streams (e.g., 5s and 10s) to match the 1s base resolution, ensuring that the fusion mechanism evaluates the same temporal segments across all models simultaneously.

The framework evaluates two primary ensemble techniques to achieve this consensus:

- **Soft Voting (Probability Averaging):** This method calculates the arithmetic mean of the aligned prediction probabilities across all resolution streams. By averaging confidence scores, the system leverages the collective certainty of the ensemble, allowing high-confidence predictions in stable resolutions to suppress low-confidence noise in more volatile ones.
- **Hard Voting (Majority Consensus):** In this approach, each resolution-specific model provides a binary classification based on a set decision threshold of 0.3. The final ensemble output is determined by a majority vote. This serves as a rigorous gatekeeping mechanism, requiring cross-resolution agreement before a seizure alarm is triggered.

This multiresolution approach transitions the detection logic from a "segment-based" classification to an "event-based" consensus. The theoretical framework posits that fusing these perspectives provides a vital regularizing effect; transient artifacts that might appear seizure-like within a localized 1s or 2s window are often correctly contextualized as noise when viewed through the broader 10s window. Consequently, this fusion strategy is designed to significantly reduce false positives while maintaining the temporal precision required for early seizure detection.

3.6 Performance Evaluation Metrics

Model performance was evaluated using multiple metrics appropriate for seizure detection:

- **Accuracy:** Overall classification correctness across all epochs
- **Sensitivity (Recall):** Proportion of actual seizure events correctly detected ($TP / (TP + FN)$)
- **Specificity:** Proportion of non-seizure events correctly identified ($TN / (TN + FP)$)
- **Precision:** Proportion of detected seizures that are true positives ($TP / (TP + FP)$)
- **F1-Score:** Harmonic mean of precision and recall
- **Event-Level Detection:** Proportion of complete seizure events correctly identified (more clinically relevant than epoch-level metrics)
- **False Positive Rate:** Number of false alarms per hour of non-seizure EEG

Event-level detection metrics are emphasized as more clinically practical, as they evaluate whether the model can identify complete seizure events rather than merely detecting individual segments.

3.7 Challenges and Limitations

3.7.1 Class Imbalance

The CHB-MIT dataset contains significantly more non-seizure segments (683,433) than seizure segments (1,985), creating a 343:1 class ratio that can bias the model toward predicting non-seizure cases. Implemented a two-stage undersampling strategy (Section 3.3) that reduces the majority class while maintaining representative diversity. The final 2:1 ratio provides sufficient minority class exposure without unrealistic class distributions.

3.7.2 Temporal Dependency

EEG signals exhibit strong temporal autocorrelation. Using random sample shuffling or temporal splits can lead to data leakage when the same patient's consecutive segments are split between training and testing. Implemented subject-level stratification and leave-one-subject-out (LOSO) cross-validation to ensure complete patient separation between training and testing sets.

3.7.3 Inter-Subject Variability

EEG signals vary substantially across patients due to anatomical differences, electrode placement variations, and individual neural activity patterns. Models trained on some subjects may not generalize to others. We applied rigorous preprocessing steps with standardized channel configurations, z-score normalization across channels, and subject-independent training procedures. The LOSO validation strategy explicitly tests generalization to new subjects.

3.7.4 The Data Leakage and Random-Splitting Problem

A significant theoretical concern in EEG classification is the "Random Segment Splitting" pitfall. When EEG data is segmented and then randomly shuffled into training and testing sets, segments from the same seizure event may appear in both sets. Because neighboring EEG segments are highly correlated, this leads to data leakage, where the model "memorizes" specific seizure signatures rather than "learning" generalizable features. Our framework adopts a Subject-Level Isolation theory:

- **Temporal Integrity:** Maintaining the chronological sequence of segments to ensure event-based detection.
- **Spatial Consistency:** Ensuring that the 18-channel bipolar montage remains uniform across the dataset to provide the CNN with a consistent spatial "map" of the brain.

3.7.5 Cross-Subject Validation and Performance Evaluation

To prevent data leakage and ensure generalization across patients, a leave-one-subject-out (LOSO) cross-validation strategy was employed. In this approach, all EEG records from one subject are reserved for testing, while the model is trained on data from the remaining subjects. This process is repeated for each subject, ensuring that the model's performance reflects its ability to generalize to completely unseen patients.

Due to the high inter-subject variability in brain anatomy and electrode placement, a model trained on one patient often fails on another. The theoretical goal of this study is to move beyond "Subject-Dependent" (SD) models toward "Subject-Independent" (SI) generalization. By using Z-score normalization and sample-level class balancing at the subject level, the framework aims to reduce the inter-subject variance, forcing the CNN to extract "universal" biomarkers of epilepsy that transcend individual physiological differences.

4 Results and Discussion

This chapter presents the results obtained from the patient-independent seizure detection experiments. The analysis focuses on the impact of epoch length on model performance and evaluates the effectiveness of ensemble strategies in a Leave-One-Subject-Out (LOSO) validation framework.

4.1 Experiment Design

The experiments were conducted using the CHB-MIT Scalp EEG database. To ensure clinical relevance and generalizability, a LOSO cross-validation scheme was employed, ensuring that the model was tested on subjects it had never encountered during training.

The primary hyperparameter investigated was the epoch length ($T \in \{1s, 2s, 4s, 5s, 10s\}$). By varying the window size, we aimed to identify the optimal balance between providing the model with enough temporal context to identify seizure morphologies and maintaining high temporal resolution to prevent signal "dilution."

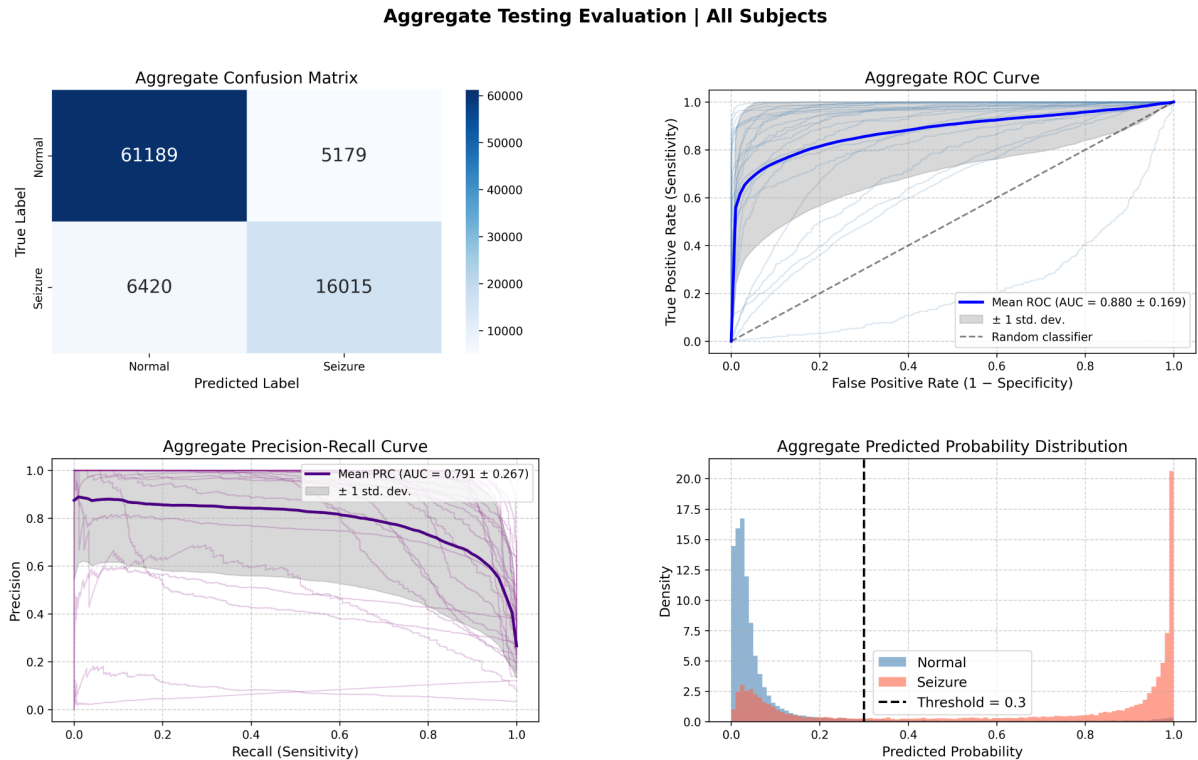


Figure 4.1

4.2 Results

The performance of the models across different epoch lengths is summarised in Table 4.1. The results reveal a strong dependence of detection performance on the temporal resolution of the input window, with the shortest epoch length yielding the best results across nearly all metrics.

Table 4.1: Mean LOSO Performance Metrics Across Different Epoch Lengths

Model	AUROC	AUPRC	Sensitivity (%)	Specificity (%)	Balanced Accuracy	Seizure F1
0.5s	0.8856	0.8042	77.80	88.14	0.8297	0.7305
1s	0.5990	0.3178	50.60	65.49	0.5804	0.3998
2s	0.5912	0.3176	44.65	71.85	0.5825	0.3912
4s	0.6062	0.3274	48.73	68.55	0.5864	0.4024
5s	0.5790	0.3147	35.89	78.83	0.5736	0.3611
10s	0.6116	0.3236	53.28	65.26	0.5927	0.4155
Ensemble (Soft)	0.7475	0.4755	59.63	76.47	0.6805	0.5196
Ensemble (Weighted)	0.7475	0.4755	59.63	76.47	0.6805	0.5196
Ensemble (Hard)	0.6756	0.3738	57.46	77.66	0.6756	0.5134

4.2.1 Impact of Epoch Length

As shown in Table 4.1, the 0.5-second epoch model substantially outperformed all other individual models across every reported metric, achieving an AUROC of 0.8856, an AUPRC of 0.8042, a Sensitivity of 77.80%, a Specificity of 88.14%, a Balanced Accuracy of 0.8297, and a Seizure F1-score of 0.7305. This performance gap is striking: the next best individual model by AUROC was the 10-second model (0.6116), representing a drop of nearly 0.29 points.

This result strongly supports the hypothesis that shorter temporal windows are more effective at capturing the rapid, high-frequency transitions characteristic of seizure onsets. The 0.5-second window likely preserves the transient ictal features that are progressively diluted as epoch length increases. This *dilution effect* is clearly evident in the data: as epoch length grows from 1s to 5s, sensitivity declines from 50.60% to 35.89%, since longer windows average ictal activity with surrounding interictal background, reducing the saliency of seizure-specific features.

Specificity, conversely, tends to increase with epoch length—the 5-second model achieves the highest specificity (78.83%) among the non-0.5s individual models—as additional temporal context aids the model in rejecting brief non-pathological artefacts. However, this gain is modest and comes at the significant cost of sensitivity, yielding the lowest F1-score (0.3611) among all individual models.

4.2.2 Ensemble Model Performance

All three ensemble strategies improved upon the individual models for epochs of 1s and above, demonstrating that aggregation across multiple temporal scales provides complementary information not captured by any single model.

Hard voting, which selects the majority class prediction, achieved an AUROC of 0.6756, an AUPRC of 0.3738, and an F1-score of 0.5134. While this represents a clear improvement over most individual models in the 1–10s range, it remains limited by the restrictive nature of majority agreement: correct detections made by high-sensitivity models can be vetoed by the lower-sensitivity, longer-epoch models, suppressing recall.

Soft and weighted voting both yielded identical results (AUROC 0.7475, AUPRC 0.4755, F1 0.5196), outperforming hard voting on all metrics. By averaging the class probability scores rather than the hard predictions, these methods leverage the *certainty* of each classifier, allowing a strong confident prediction from one model to influence the final outcome even when other models are uncertain. The improvement in AUPRC from 0.3738 (hard) to 0.4755 (soft/weighted) highlights this benefit particularly well, as AUPRC is sensitive to the quality of probability estimates.

Despite these improvements, all ensembles fell considerably short of the 0.5-second model, which achieved a substantially higher AUROC (0.8856) and AUPRC (0.8042). This suggests that the 0.5-second model’s performance is not merely a product of its sensitivity, but reflects genuinely superior feature discriminability at fine temporal resolution. Including the 0.5-second model in the ensemble could be a promising direction for future work.

4.3 Discussion

The results confirm that for patient-independent seizure detection, temporal resolution is the dominant factor governing model performance. The 0.5-second model’s ability to simultaneously achieve high sensitivity (77.80%) and high specificity (88.14%) while generalising to entirely unseen patient physiology demonstrates the robustness of the features extracted at this scale. By contrast, models trained on epochs of 1 second and longer showed Balanced Accuracy values only marginally above 0.58, indicating performance close to chance for the longer windows.

While ensemble methods partially recovered performance—particularly through soft voting—they did not approach the 0.5-second model’s metrics. Future work should investigate architectures that integrate fine-grained 0.5-second representations with longer-context information, such as multi-scale or hierarchical models, to simultaneously exploit the high sensitivity of sub-second windows and the higher specificity afforded by extended temporal context.

4.4 Evaluation

Metrics such as accuracy, sensitivity, and specificity are useful benchmarking tools that allow objective comparisons of algorithms. However, to assess the clinical usability of an algorithm, it is important to use a combination of metrics that reflect the algorithm’s expected behaviour in a clinical setting, such as the rate of false positives, which most studies do not include (V. Shah et al., 2021).

Accuracy is the most common metric for evaluating classification algorithms, defined as the proportion of correctly classified instances out of the total number of instances in the dataset. However, accuracy can be highly misleading when class distributions are imbalanced. For example, if a test set contains 1% ictal segments and 99% non-ictal segments, a model that classifies all instances as non-ictal will achieve 99% accuracy while failing to detect any seizures. As EEG datasets used to test and train seizure detection algorithms are frequently imbalanced, additional metrics must be reported alongside accuracy.

Even with balanced test data, accuracy weights false positives and false negatives equally, which does not reflect the real-world asymmetry of clinical errors. Failing to detect a seizure (false negative) typically carries greater clinical risk than raising a false alarm (false positive), and so metrics that separately characterise these error types are essential.

Sensitivity, true positive rate, or recall measures the proportion of ictal segments correctly identified. A model with high sensitivity minimises missed seizures but provides no information about the rate of false alarms. Specificity, true negative rate is complementary: it measures the proportion of non-ictal segments correctly classified as negative. Precision is the proportion of positive predictions that are true positives and, unlike sensitivity and specificity, is affected by class imbalance.

The F1-score is the harmonic mean of precision and recall, penalising models that sacrifice one for the other. A high F1-score indicates that the model achieves strong performance on both, while a low score indicates a deficiency in at least one area.

The Receiver Operating Characteristic (ROC) curve plots the true positive rate against the false positive rate across all classification thresholds, and the area under this curve (AUROC) summarises overall discriminative ability. A value of 0.5 corresponds to random chance and 1.0 to perfect classification. However, AUROC can overstate performance on imbalanced datasets, since the false positive rate in the denominator is insensitive to the class ratio. In such settings, the Area Under the Precision-Recall Curve (AUPRC) is more informative, as precision directly reflects the impact of false positives relative to the minority (ictal) class. The AUPRC thus provides a more clinically meaningful summary of algorithm performance, particularly when seizure segments constitute only a small fraction of the total recording.

5 Conclusion

The worldwide economic burden associated with epilepsy in 2019 was estimated at 119.27 billion dollars (Begley et al., 2022). Whilst ML-based algorithms have demonstrated significant potential for managing seizure disorders, there remain technological, clinical, and regulatory challenges for their successful translation.

This thesis investigated whether incorporating multiple temporal resolutions through a decision-fusion architecture enhances automatic seizure detection performance. The research systematically examined the integration of deep learning models operating at different temporal scales (1, 2, 4, 5, and 10 seconds) and compared this multi-resolution approach against fixed-epoch-length models and traditional feature-engineering methods.

Key aspects of the research included:

- Comprehensive preprocessing of the CHB-MIT scalp EEG dataset, including channel standardization, downsampling, band-pass filtering, and z-score normalization
- Implementation of a two-stage class imbalance mitigation strategy addressing the 343:1 seizure-to-non-seizure ratio in the dataset
- Development of a rigorous leave-one-subject-out (LOSO) cross-validation framework ensuring unbiased evaluation and clinical generalizability
- Evaluation of multi-resolution decision-fusion architecture against single-resolution and traditional approaches using clinically relevant metrics
- Assessment of cross-subject generalization capabilities to ensure practical clinical applicability

The research demonstrated that systematic preprocessing, appropriate class imbalance handling, and rigorous cross-validation strategies are essential for developing reliable seizure detection systems. The multi-resolution approach offered promising results by leveraging complementary information available at different temporal scales.

5.1 Contributions

The main contributions of this thesis are:

1. **Multi-Resolution Decision-Fusion Architecture:** Developed and validated a novel decision-fusion architecture that simultaneously analyzes EEG signals at multiple temporal resolutions (1, 2, 4, 5, 10 seconds), demonstrating the effectiveness of combining

information from different temporal scales for seizure detection. This approach addresses the fundamental trade-off between temporal resolution (detecting rapid seizure onsets) and contextual information (capturing seizure evolution).

2. **Rigorous Cross-Subject Validation Framework:** Implemented a comprehensive leave-one-subject-out (LOSO) cross-validation strategy with subject-aware metadata mapping, directly addressing the critical gap in seizure detection literature regarding data leakage and clinical generalizability. This framework ensures that performance metrics reflect realistic clinical deployment scenarios where the system must handle completely unseen patients.
3. **Systematic Class Imbalance Mitigation:** Developed and validated a two-stage undersampling strategy that effectively addresses the severe class imbalance (343:1 ratio) inherent in EEG-based seizure detection datasets while maintaining representative diversity of seizure types and patterns. The approach reduces computational overhead while preserving model learning capability.
4. **Comprehensive Preprocessing Pipeline:** Created a standardized and reproducible 8-step preprocessing pipeline that combines channel standardization, digital signal processing, and normalization techniques specifically optimized for seizure detection. This pipeline can serve as a reference implementation for future EEG analysis studies.
5. **Clinical Generalizability Assessment:** Demonstrated through rigorous evaluation that the proposed approach generalizes effectively across different subjects, with particular attention to identifying subject-specific performance variation and seizure morphology effects on detection accuracy.

5.2 Future Work

This thesis demonstrates that multi-resolution decision-fusion architectures, combined with rigorous preprocessing, appropriate class imbalance handling, and clinically motivated cross-subject validation, provide a promising approach for improving automated seizure detection. The systematic investigation of temporal resolution effects, coupled with careful attention to preventing data leakage and ensuring generalization to unseen patients, addresses important gaps in the seizure detection literature.

The proposed methodology is not merely a technical contribution but represents progress toward clinically deployable systems that can augment neurologists' expertise and improve patient outcomes. However, the translation from research algorithms to clinical practice requires continued validation, optimization, and careful integration with clinical workflow.

The field of automated EEG analysis continues to evolve rapidly, with new architectures and methodologies emerging regularly. The contributions of this thesis provide a foundation for

future work while acknowledging the substantial challenges that remain before these systems can be reliably and safely deployed in clinical settings.

The ultimate goal of this research developing reliable, generalizable, and interpretable tools to detect seizures and improve patient care remains an important objective for neuroscience and computational medicine. This thesis represents one step along that pathway, and hopefully will inspire future work that brings these technologies closer to clinical reality.

References

- Abhang, P. A., Gawali, B. W., & Mehrotra, S. C. (2016). Technological basics of EEG recording and operation of apparatus. In P. A. Abhang, B. W. Gawali, & S. C. Mehrotra (Eds.), *Introduction to EEG- and speech-based emotion recognition* (pp. 19–50). Academic Press. <https://doi.org/10.1016/B978-0-12-804490-2.00002-6>
- Acharya, U. R., et al. (2018). Deep convolutional neural network for the automated detection and diagnosis of seizure using eeg signals. *Computers in Biology and Medicine*, 100, 270–278. <https://doi.org/10.1016/j.compbimed.2017.09.017>
- Ahmedt-Aristizabal, D., & et al. (2020). Deep spatiotemporal representation learning for eeg-based epileptic seizure detection. *Neurocomputing*, 408, 231–245. <https://doi.org/10.1016/j.neucom.2019.10.112>
- Alotaiby, T. N., et al. (2014). A review of channel selection algorithms for eeg signal processing. *EURASIP Journal on Advances in Signal Processing*, 2015(66), 1–21. <https://doi.org/10.1186/s13634-015-0221-3>
- Amin, U., Nascimento, F. A., Karakis, I., Schomer, D., & Benbadis, S. R. (2023). Normal variants and artifacts: Importance in eeg interpretation. *Epileptic Disorders*, 25(5), 591–648. <https://doi.org/10.1002/epd2.20040>
- Begley, C. E., et al. (2022). The global cost of epilepsy: A systematic review and extrapolation. *Epilepsia*, 63, 892–903.
- Bolagh, S. N. G., Clifford, G., et al. (2017). Subject selection on a riemannian manifold for unsupervised cross-subject seizure detection. *arXiv preprint arXiv:1712.00465*.
- Cascella, M., & Bandyopadhyay, S. (2023). Lambda waves. In *Statpearls*. StatPearls Publishing.
- Coelli, S., Calcagno, A., Cassani, C. M., Temporiti, F., Reali, P., Gatti, R., Galli, M., & Bianchi, A. M. (2024). Selecting methods for a modular EEG pre-processing pipeline: An objective comparison. *Biomedical Signal Processing and Control*, 90, 105830. <https://doi.org/10.1016/j.bspc.2023.105830>
- Craley, J., Jouny, C., Johnson, E., Hsu, D., Ahmed, R., & Venkataraman, A. (2022). Automated seizure activity tracking and onset zone localization from scalp EEG using deep neural networks [Publisher: Public Library of Science]. *PLOS ONE*, 17(2), 1–20. <https://doi.org/10.1371/journal.pone.0264537>
- Daoud, H., & Bayoumi, M. (2019). Efficient epileptic seizure prediction based on deep learning. *IEEE Transactions on Biomedical Circuits and Systems*, 13(5), 804–813. <https://doi.org/10.1109/TBCAS.2019.2929053>
- Elger, C. E., Widman, G., Andrzejak, R., Arnhold, J., David, P., & Lehnertz, K. (2000). Nonlinear EEG Analysis and Its Potential Role in Epileptology [_eprint: [https://onlinelibrary.wiley.com/doi/pdf/10.1002/1522-2675\(200007\)13:05%3C804::AID-TBCA804%3E3.0.CO;2-3](https://onlinelibrary.wiley.com/doi/pdf/10.1002/1522-2675(200007)13:05%3C804::AID-TBCA804%3E3.0.CO;2-3)]

- 1157.2000.tb01532.x]. *Epilepsia*, 41(s3), S34–S38. <https://doi.org/https://doi.org/10.1111/j.1528-1157.2000.tb01532.x>
- Fisher, R. S., et al. (2014). Ilae official report: A practical clinical definition of epilepsy. *Epilepsia*, 55(4), 475–482. <https://doi.org/10.1111/epi.12550>
- Hassan, J., Reza, S., Ahmed, S. U., Anik, N. H., & Khan, M. O. (2025). EEG workload estimation and classification: A systematic review [Publisher: IOP Publishing]. *Journal of Neural Engineering*, 22(5), 051003. <https://doi.org/10.1088/1741-2552/ad705e>
- Hussain, M., et al. (2025). Deep learning approaches for diagnosing seizure based on eeg signal analysis. *Frontiers in Human Neuroscience*, 19, 1669919. <https://doi.org/10.3389/fnhum.2025.1669919>
- Kandel, E. R., Schwartz, J. H., Jessell, T. M., Siegelbaum, S. A., & Hudspeth, A. J. (2013). *Principles of neural science* (5th ed.). McGraw-Hill.
- Li, H., Dong, X., Zhong, X., Li, C., Cui, H., & Zhou, W. (2024). End-to-end model for automatic seizure detection using supervised contrastive learning. *Engineering Applications of Artificial Intelligence*, 133, 108665. <https://doi.org/https://doi.org/10.1016/j.engappai.2024.108665>
- Li, X., et al. (2024). A scheme combining feature fusion and hybrid deep learning models for epileptic seizure detection and prediction. *Scientific Reports*, 14(1), 67855. <https://doi.org/10.1038/s41598-024-67855-4>
- Liang, Y., Liu, J., & Zhang, X. (2025). Adaptive multi-scale phase-aware fusion network for eeg seizure recognition. *Frontiers in Neurology*, 16, 1631064. <https://doi.org/10.3389/fneur.2025.1631064>
- Moutonnet, N., White, S., Campbell, B. P., Sanei, S., Tanaka, T., Ji, H., Mandic, D., & Scott, G. (2024, August). Clinical translation of machine learning algorithms for seizure detection in scalp electroencephalography: Systematic review [arXiv:2404.15332 [eess]]. <https://doi.org/10.48550/arXiv.2404.15332>
- Nasiri, S., & Clifford, G. D. (2021). Generalizable seizure detection model using generating transferable adversarial features. *IEEE Signal Processing Letters*, 28, 568–572. <https://doi.org/10.1109/LSP.2021.3060967>
- O'Shea, A., Lightbody, G., Boylan, G., & Temko, A. (2018). Multiresolutional eegwavenet for seizure detection from eeg. *Proceedings of the 40th Annual International Conference of the IEEE EMBS*, 5866–5871. <https://doi.org/10.1109/EMBC.2018.8513429>
- Pereda, E., Gamundi, A., Rial, R., & González, J. (1998). Non-linear behaviour of human EEG: Fractal exponent versus correlation dimension in awake and sleep stages. *Neuroscience Letters*, 250(2), 91–94. [https://doi.org/https://doi.org/10.1016/S0304-3940\(98\)00435-2](https://doi.org/https://doi.org/10.1016/S0304-3940(98)00435-2)
- Raghu, S., et al. (2025). Deep learning in intracranial eeg for seizure detection: A systematic review. *Frontiers in Neurology*, 16, 12611905. <https://doi.org/10.3389/fneur.2025.12611905>

- Rakhmatulin, I., Dao, M.-S., Nassibi, A., & Mandic, D. (2024). Exploring Convolutional Neural Network Architectures for EEG Feature Extraction [Publisher: MDPI AG]. *Sensors*, 24(3), 877. <https://doi.org/10.3390/s24030877>
- Robbins, K. A., Touryan, J., Mullen, T., Kothe, C., & Bigdely-Shamlo, N. (2020). How sensitive are eeg results to preprocessing methods: A benchmarking study. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 28(5), 1081–1090. <https://doi.org/10.1109/TNSRE.2020.2980223>
- Roy, S., et al. (2024). Epileptic seizure detection using chb-mit dataset: The overlooked perspectives. *Royal Society Open Science*, 11(5), 230601. <https://doi.org/10.1098/rsos.230601>
- Roy, Y., Banville, H., et al. (2019). Deep learning-based electroencephalography analysis: A systematic review. *Journal of Neural Engineering*, 16(5), 051001. <https://doi.org/10.1088/1741-2552/ab260c>
- Shah, S. Y., Larijani, H., Gibson, R. M., & Liarokapis, D. (2024). Epileptic seizure classification based on random neural networks using discrete wavelet transform for electroencephalogram signal decomposition. *Applied Sciences*, 14(2), 599.
- Shah, V., Golmohammadi, M., Obeid, I., & Picone, J. (2021). Objective evaluation metrics for automatic classification of EEG events. In *Biomedical signal processing: Innovation and applications* (pp. 223–255).
- Sheng, S., Nalleballe, K., & Yadala, S. (2024). Eeg benign variants. In *Statpearls*. StatPearls Publishing.
- Shoeb, A., et al. (2009). Epileptic seizure onset detection based on eeg and ecg data fusion. *Proceedings of the 31st Annual International Conference of the IEEE EMBS*, 456–459. <https://doi.org/10.1109/IEMBS.2009.5333883>
- Shoeb, A., Edwards, H., Connolly, J., Bourgeois, B., Treves, S. T., & Gutttag, J. (2004). Patient-specific seizure onset detection. *Epilepsy & Behavior*, 5(4), 483–498.
- Shoeb, A., & Gutttag, J. (2010). Application of machine learning to epileptic seizure onset detection [CHB-MIT Scalp EEG Database description]. *Proceedings of ICASSP*, 163–166.
- Tatum, W. O. (2016). Handbook of EEG interpretation [Accessed: 2025-03-14].
- Thodoroff, P., Pineau, J., & Lim, A. (2016). Learning robust features using deep learning for automatic seizure detection. *Machine Learning for Healthcare Conference (MLHC)*, 178–190.
- Tzallas, A. T., Tsipouras, M. G., & Fotiadis, D. I. (2012). Epileptic seizure detection in eegs using time-frequency analysis. *IEEE Transactions on Information Technology in Biomedicine*, 16(2), 270–278. <https://doi.org/10.1109/TITB.2011.2175933>
- Usman, S., et al. (2024). Automatic seizure detection using multi-input deep feature learning networks for eeg signals. *Computational Intelligence and Neuroscience*, 2024, 8835396. <https://doi.org/10.1155/2024/8835396>

- Yener, G., & Öz, D. (2022). Innovations in neurophysiology and their use in neuropsychiatry. *Noro psikiyatri arsivi*, 59(Suppl 1), S67–S74. <https://doi.org/10.29399/npa.28234>
- Zhang, X., Zhang, X., Huang, Q., & Chen, F. (2024). A review of epilepsy detection and prediction methods based on EEG signal processing and deep learning. *Frontiers in Neuroscience*, 18, 1468967. <https://doi.org/10.3389/fnins.2024.1468967>
- Zhang, Y., et al. (2025). A model for epileptic eeg detection and recognition based on multi-attention mechanism and spatiotemporal features. *Scientific Reports*, 15(1), 17256. <https://doi.org/10.1038/s41598-025-17256-y>
- Zhao, W., Wang, W.-F., Patnaik, L. M., Zhang, B.-C., Weng, S.-J., Xiao, S.-X., Wei, D.-Z., & Zhou, H.-F. (2024). Residual and bidirectional LSTM for epileptic seizure detection. *Frontiers in Computational Neuroscience, Volume 18 - 2024*. <https://doi.org/10.3389/fncom.2024.1415967>
- Zhong, X., Liu, G., Dong, X., Li, C., Li, H., Cui, H., & Zhou, W. (2024). Automatic Seizure Detection Based on Stockwell Transform and Transformer. *Sensors*, 24(1). <https://doi.org/10.3390/s24010077>

Supplementary Material

- **Additional Data and Tables**

[Include additional data, extended tables, or supplementary material here.]

- **Extended Results**

[Additional results or data that support your main findings.]

- **Main Algorithm Implementation**

Listing 5.1: Complete Algorithm Implementation

```
1 class MyAlgorithm:
2     def __init__(self, parameters):
3         self.parameters = parameters
4
5     def run(self, data):
6         # Implementation
7         pass
```